



SPONTOM ENTERPRISE PRIVATE LIMITED

Build with AI (Google) • Track 5

TatRaksha

**From a cyclone bulletin
to village-level action.**

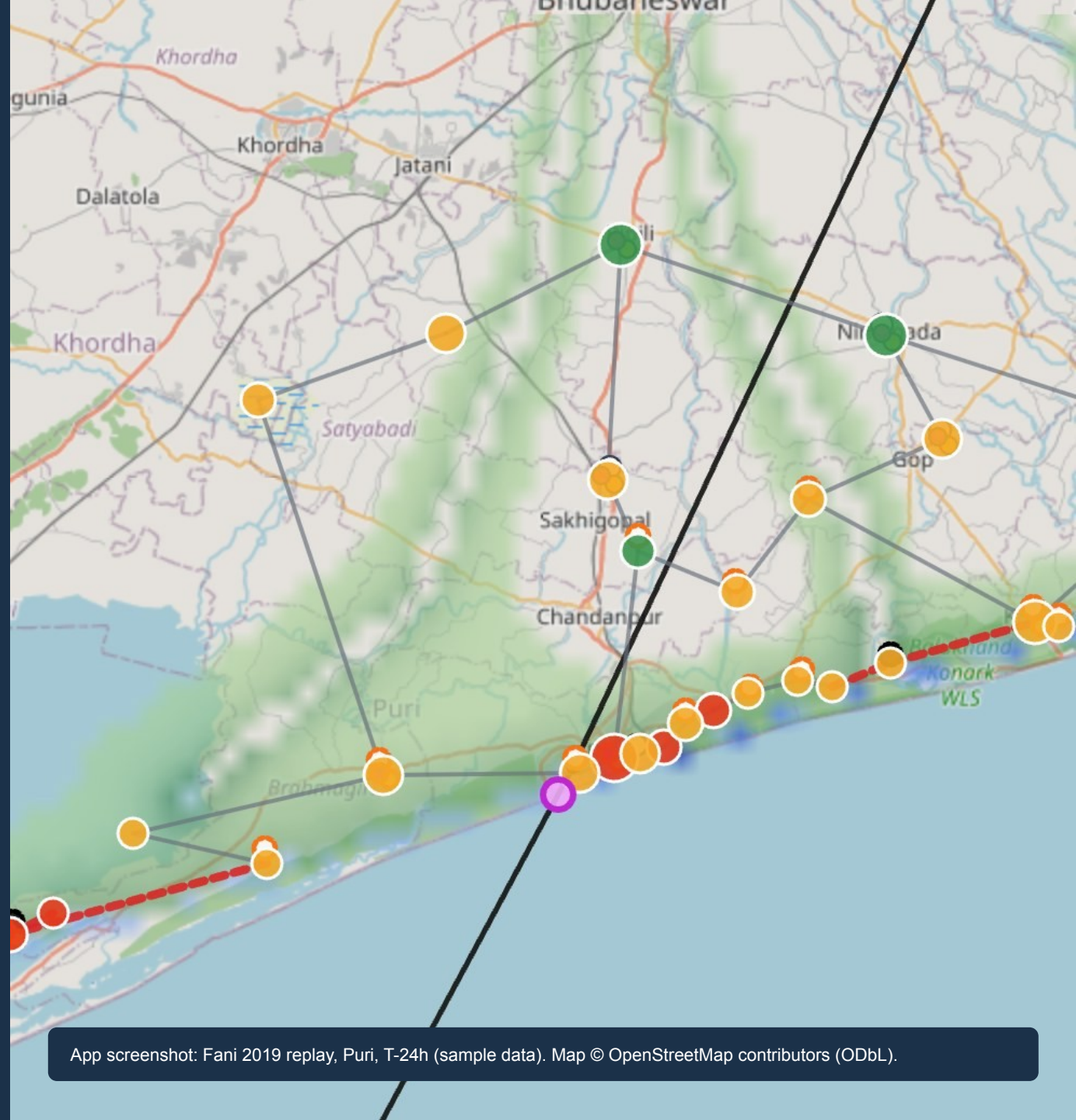
Impact forecasts, a Gemini situation report and approved multilingual warnings for Bay of Bengal districts.

GEMINI 3.7 FLASH

HUMAN-APPROVED

A prototype by Spontom Enterprise Private Limited

DPIIT Recognised • Visakhapatnam • September 2026



App screenshot: Fani 2019 replay, Puri, T-24h (sample data). Map © OpenStreetMap contributors (ODbL).

The forecast is regional. The decisions are local.

A district officer must turn each new bulletin into who to evacuate, what to protect and what to say.

One bulletin. Hundreds of local decisions.

72

hours

30

villages

1

officer

Fani replay, T-24h: 59,600 people in high-risk villages; 5 villages with no shelter reachable by road.

LOCAL-IMPACT GAP

BULLETIN

Track, wind, surge text

COMPILE

Lists, maps, phone calls

?

Who floods? Which road cuts?

WARN

Drafted by hand, per language

FOCUSED PROBLEM

Officers lack a village-level impact picture that updates with every bulletin.

Consequence Evacuation is late or mistargeted, shelters turn out unreachable, warnings lag in local languages.

[1] IMD RSMC bulletins give regional track/surge guidance. [7] Replay figures: TatRaksha Fani 2019 sample data (synthetic terrain), not official statistics.

TatRaksha turns each bulletin into an approved warning.

One pipeline from the official forecast to a CAP 1.2 alert, with a human decision at the end.

01

IMPACT FORECAST

Surge, rain-flood and wind give every village a transparent Village Risk score.

02

GEMINI REASONING

Multimodal situation report from hazard map, bulletin and tables; numbers checked.

03

APPROVED WARNING

Odia, Bengali or English advisory; officer approves; CAP 1.2; sandbox dispatch.

ADVISORY DRAFT • SATAPADA

NEEDS APPROVAL

Evacuation order • Fani • T-24h • English

1

What: winds up to 170 km/h, heavy rain and flooding

2

Where: Satapada, Krushnaprasad block, Puri

3

When: from 02-05-2019 17:27 IST

4

Action: Leave now for the cyclone shelter. Do not wait.

5

No shelter reachable by road: strong building on high ground; call 1070.

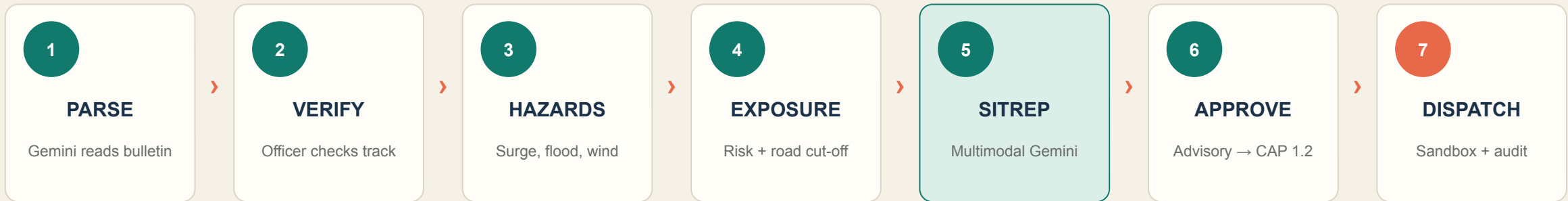
Issued by Collector & DM, Puri (DDMA)

APPROVE → CAP

Real output of the app (advisory_v1, gemini-3.7-flash) on the Fani 2019 replay at T-24h; replay data are samples. [4] OASIS CAP 1.2.

Every new bulletin re-runs the whole chain.

Observed data, model estimates, AI interpretation and recommendation stay separate and labelled.



AUTO • PER BULLETIN

Parse, 9-track set, hazards, Village Risk, cut-offs and sitrep draft re-run on each update.

OFFICER • DECIDES

Verifies the track, reviews translations, approves or edits each advisory before dispatch.

STATE EOC • COMPARES

Sees districts side by side, advisory logs and parametric trigger estimates.

HUMAN GATE

No advisory is dispatched without the district officer's approval; parametric numbers are planning estimates.

Journey per PRD §8. Warning stages follow the PRD: T-72h Watch → T-48h Warning → T-24h Evacuation Order → T-6h Final Warning.

Built for the district officer, useful to the state.

Two app roles only. Everyone else receives role-specific actions or public advisories.

Substations exposed
5

Hospitals exposed
4

Villages without reachable shelter
5

Shelters cut off
2

Shelters exposed
15

Actions by role

District Authority

Execute mandatory evacuation orders for 59,600 people in high-risk zones and relocate evacuees from unreachable/over-capacity shelter areas to secondary safe sites.
Evidence: $\text{people_high_risk} = 59600 \cdot \text{evacuation_demand} = 25350 \cdot \text{villages_without_reachable_shelter} = 5$

by T-12h

Power Utility

Implement precautionary shutdown and de-energization protocols for exposed substations subject to high wind and inundation.

by T-8h

DISTRICT OFFICER • PRIMARY USER

Gemini sitrep: actions by role, with deadlines and evidence

01

STATE EOC OFFICER

Secondary user

Compares districts, reviews advisories, sees parametric triggers.

02

COLLECTOR & DM

Issuing authority

Named on every advisory; approval runs through the DDMA.

03

LINE DEPARTMENTS

Action owners

Power, health, police, fisheries get tasks with deadlines, e.g. power by T-8h.

04

COASTAL PUBLIC

Recipients

Fishers and villagers get warnings in Odia, Bengali or English; no app login.

CONSTRAINT → DESIGN RESPONSE

UPDATES

re-run per bulletin

LANGUAGES

templates + voice

DATA GAPS

labels + freshness

Roles from PRD §7. Screenshot: live sitrep on the Fani replay (gemini-3.7-flash, sitrep_v1); figures are from sample data.

Fani 2019 replay: Satapada at T-24 hours.

Scenario A of PRD §43: Odisha, Puri district, Odia, storm surge plus wind. All figures from sample replay data.



GEMINI SITUATION REPORT • T-24H

+

59,600 people in high-risk villages

✓

Power: 5 substations by T-8h

↗

Penthakata: 23,600 assigned, 1,650 seats

Evacuate by T-12h.

Evidence-linked actions:

WHY TRUST IT

Numbers checked against the pipeline (grounded: true); model, version and prompt stored with the report.

Sample data on the synthetic terrain grid; surge is a screening estimate – defer to official surge guidance. Captured from the app, 30 Sep 2026.

Not a new forecast. The missing local-impact layer.

India's systems forecast and disseminate well; TatRaksha connects the forecast to village-level decisions.

APPROACH	WHAT IT CONTRIBUTES	REMAINING BOUNDARY / GAP
IMD RSMC bulletins	Authoritative track, intensity, surge guidance	Coast/district scale; local impact is compiled by hand
INCOIS ocean services	Storm-surge and high-wave forecasts for the coast	Hazard, not people, shelters or roads
NDMA SACHET (CAP)	Standard, geo-targeted alert dissemination	Delivers a message; does not decide who or what
Generic AI assistant	Fast drafting and summarising	No grounding, approval gate or audit trail
TatRaksha	Village risk + road cut-off + grounded Gemini sitrep + CAP	Screening models; needs field validation; unproven

DISTINCTIVE CAPABILITY

Feeds official forecasts in and CAP alerts out: it complements, not replaces, national systems.

[1] IMD RSMC New Delhi. [2] INCOIS. [3] NDMA SACHET. TatRaksha consumes official forecasts and defers to official surge guidance.

Bounded Gemini on transparent hazard models.

Gemini reads and reasons; it never measures. Every AI output is schema-validated JSON.

01	OFFICIAL INPUTS	IMD bulletin text/PDF • state adapter JSON • canonical schema
02	HAZARD ENGINE	Screening surge • rain-flood heuristic • Rankine wind • 9-track set
03	EXPOSURE + ROUTING	Village Risk 30/25/15/15/15 • road cut-off • Google Routes
04	GEMINI LAYER	Gemini 3.7 Flash: parse, multimodal sitrep, drafts; versioned prompts
05	DELIVERY + AUDIT	CAP 1.2 • TTS/STT • sandbox dispatch • BigQuery audit • Cloud Run

GOOGLE AI INTEGRATION MAP

● Live: Gemini 3.7 Flash on Vertex AI

● Live: Maps Routes, Translation, TTS, STT

● Live: BigQuery audit, Cloud Storage archive

NEXT

Earth Engine layers wired; Vertex AI flood model.

REAL VS DEMO, AND FRESHNESS

Startup probe

One live call per integration; the badge shows what truly works

Sample flag

Replay data carry is_sample in the data and the UI

Stale input

A newer bulletin marks the analysis out of date

[6] Earth Engine data catalog. Rain-flood likelihood is a transparent heuristic until the Vertex AI model is trained (PRD §12).

Life-safety AI must be checkable and overridable.

Prefer caution, show uncertainty, and keep a human decision on every warning.

HUMAN APPROVAL

EVERY ADVISORY, EVERY DISPATCH

Grounded numbers Sitrep figures checked against the pipeline; ungrounded items removed and listed.

Provenance model_name, model_version and prompt_version stored on every AI record.

DO NOT DO / COLLECT

- Invent forecast, surge or exposure numbers
- Store citizens' phone numbers (gateways hold them)
- Send real public alerts from the MVP (sandbox only)

SAFETY BY DESIGN

SCREENING LABEL Surge is shown as a screening estimate; defer to official guidance.

UNCERTAINTY Track cone, expected vs high surge, and listed uncertainties.

WORDING Odia/Bengali from pre-approved templates; back-translation for review.

MANDATORY FIELDS What, where, when, action, shelter, authority, or rejected.

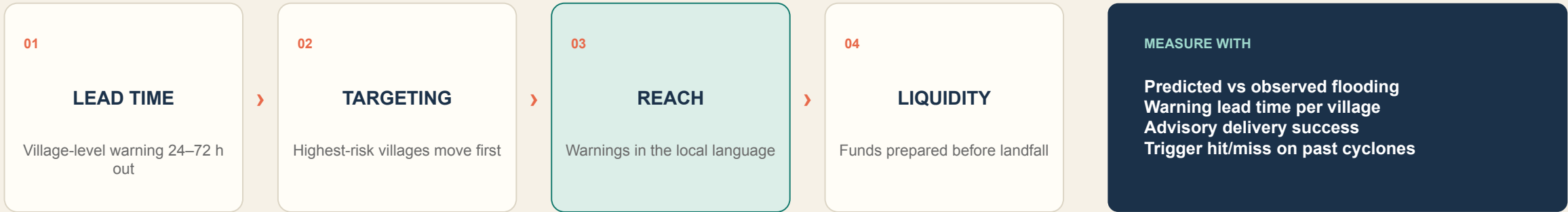
ACCESS Two roles; district officers act only on their own district.

AUDIT Append-only audit and dispatch logs streamed to BigQuery.

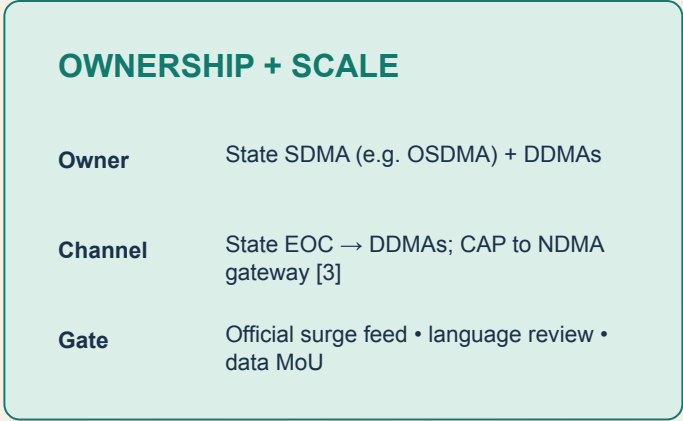
Rules from PRD §35 and §52. [4] OASIS CAP 1.2. Odia/Bengali templates are sample translations pending native-speaker review.

Prove it in shadow mode before anyone relies on it.

The first proof is earlier, better-targeted action, not app usage.



VALIDATION GATES



BUDGET ROUTE

Candidate: SDMF / SDRF preparedness and capacity-building windows under XV-FC guidelines; confirm with the SDMA. [8]

Phases per PRD §50: district pilot → state → all coastal states → Bay of Bengal (Bangladesh, Myanmar, Sri Lanka). [5] NCRMP shelters and warning systems.

Forecast, alerting and programme sources

Numbered references correspond to citations in the core slides.

[1]	IMD RSMC New Delhi	Tropical cyclone bulletins and reports, including Fani (2019) and Amphan (2020); used to reconstruct the sample replays.	OFFICIAL SOURCE ↗
[2]	INCOIS	Indian National Centre for Ocean Information Services: storm-surge and high-wave services for India's coast.	OFFICIAL SOURCE ↗
[3]	NDMA SACHET	National Disaster Management Authority's CAP-based integrated alert system for geo-targeted dissemination.	OFFICIAL SOURCE ↗
[4]	OASIS CAP v1.2	Common Alerting Protocol 1.2 standard; TatRaksha generates and validates CAP 1.2 messages.	STANDARD ↗

Programme, data and funding sources

Numbered references correspond to citations in the core slides.

[5]	NCRMP (NDMA)	National Cyclone Risk Mitigation Project: multipurpose cyclone shelters and early-warning dissemination in coastal states.	OFFICIAL SOURCE ↗
[6]	Earth Engine data catalog	Copernicus DEM, GPM IMERG rainfall, JRC Surface Water, WorldPop, Open Buildings layers used by the Earth Engine path.	DATASETS ↗
[7]	TatRaksha replay data (sample)	data/replays/: reconstructed tracks and bulletins, synthetic terrain, sample villages, shelters, roads; flagged is_sample.	LOCAL EVIDENCE
[8]	MHA Disaster Management Division	SDRF / SDMF guidelines under the XV Finance Commission award; candidate budget route for a state pilot.	OFFICIAL SOURCE ↗